Zoo-507 All past solved paper

507 Solved Past Papers

Q: Write down the component of ecology?

Two main component of ecosystem are:

i- Abiotic: non living elements: air, current, minerals

ii- Biotic: Living part of ecosystem: animal, plants, fungi, bacteria

iii- Consumers in chain

iv- Decomposers & Nutrient cycle-

Q: Sub-fields of biology?

Chemistry:- it is the study of ~~matter~~ chemicals-

Sub fields:- Ecology:- organisms interact with one an other & with their non-living things.

Palentology:- study of fossils & ancient life

Anthropology:- Study of human culture

Hydrology:- Study of earth resource i.e: lake, oceans

Climatology:- earth's atmosphere & its climatical changes

Botany:- plants Zoology:- Animals

Q3: Define Environmental Sciences?

A: It is an interdisplinary study of connection b/w earth's life support system & human activities-

Q4: Define Environmentalism?

a social movement dedicated to protecting the earth's life support system for us & for others-

Q5: Environmental Sciences fields?

Anthropology e

Demography e

Economics e

Political Sciences e

Humanities e

(ڈیڑھ)

Q: Environmental Sustainability ?

A: Sustainability is the ability of earths various natural system & human cultural system & economics to survive & adopt the changing environmental condition indefinetly.

Q: Define Sustainability ?

A cultural component of sustainability is natural resources & natural service that keep us & other forms of life alive & support our economic -

Q: Define Nutrient cycling ?

A vital service is nutrient cycling the circulation of chemical necessary for life life form, environment (mostly from air & soil) through organism & back to environment.

Q: Define & example of prokaryotes ?

They are one-celled organism that do not hv a nucleus or any of the membrane-bound organelles inside them. eg bacteria & cyanobacteria

Q: How Solar/light energy to chemical energy by plants ?

Solar energy converted to chemical energy by plants is known as photosynthesis-

-> The process by which green plants & some other organism use sunlight to synthesize nutrient from CO_2 & water.

Q:11 How green plants uses solar energy?
In plants, photosynthesis involves the green pigment chlorophyll & generates oxygen as by-product.

Q:12 Types of energy?
light energy, heat energy, gravitational ... chemical energy, sound, chemical. atomic energy. Each form can be converted or changed into other form.

Q:13 Base of energy quality?
Energy quality is contrast b/w diff forms of energy, the diff trophic levels in ecological system & the propensity of energy to convert from one to another.

Q:14 What is natural capital?
It is the world's stock of natural resources, which includes geology, soil, air, water & all living organism. that human derive a wide range of services, often called ecosystem services which makes human life possible.

Q:15 Natural capital uses & degradation?
1- It is the environmental stock of resources of Earth that provide goods, 2- flows, 3- ecological services required to support life
4- It has financial value as the use of natural 5 derive many bussines.
5- It is the better use & care of natural resources.

Q.16 6) Degradation: (natural capital degradation) involves using resources at an ~~unsu~~ unsustainable rate. Means, They are wasting the natural resources like burning of coal at extreme level.

Q.17 Effects of human on our environment?

Human impact on our environment or anthropogenic impact on the environment includes changes to biophysical environment and ecosystem, biodiversity loss, and natural resources caused directly or indirectly by humans, including global warming, environmental degradation, mass extinction of resources, over consumption, over exploitation, pollution and deforestation. These are all the result from human activity.

Q.18 How ecological footprints are growing?

The world-average ecological footprint in 2014 was 2.8 global hectares per person. The carbon footprint is the fastest growing part of ecological footprint and accounts currently for about 60% of humanity's total ecological footprint.

Imp of atmosphere
1. its very small fraction weigh on earth
3. Its imp bcz its highly reactive
4. Circulates & Plays imp role in evolution of landscape
2. The mixture of gases that surrounds earth called atmosphere

Q.19 In list composition of atmosphere in ecosystem?

4. Nitrogen accounts for 78% of atmosphere Oxygen 21 & Argon 0.9%, Gasses like carbon-dioxide, nitrous oxide, methane and ozone are traces gasses that about the tenth of one percent of atmosphere.

Q.20. Factor assembly community?

It describes how the first species to colonize an ecosystem determine what end state of that ecosystem look like. The accumulation of specie diversity over time in habitat.

Q21: Explain non renewable environment?
Cultural eutrophication is the process that speeds up natural eutrophication bcz of human activity. Due to clearing of land & buildings of towns & cities. Land runoff is accelerated & more nutrient such as phosphate & nitrate is applied to lakes & rivers, then to coastal estuaries & bays.

Q22: Why water H_2O is Important?
water is one of the most imp substance on earth. All plants & animal must have water to survive, If there is no water there would be no life on earth. Apart from drinking to survive it is use for many other works - It keeps body temprature normal, lubricate joints.

Q23: Composition of geosphere?
The three main layer of earth includes the thin outer (crust 1% of the earth's volume) the third (mostly rocks) mantle (84%) & the core (inner & outer combined, 15%)

Q24: Major factors that support life on earth?
Earth has basic requirement are a source of energy - protection from UV, water (liquid solvent), gases Ideally placed in our solar system & glory to support life.

Q25: Five example of tertary consumers?
Tertiary consumers can be carnivores.
They can eat meat of other carnivors - hawk, wolves, Tiger, leopard, snakes
Basically, they are predators feeds on primary consumers.

Q26: Five example of Secondry consumers?
feed on herbivores: cows, zebra, Rabbit, Deers.

Q:26 Imp of diatoms? Mark ND factors of diatom?
Unicellular algae, shells having interlocking pasts Major producer of aquatic system

Dinoflagellate:- unicellular, double flagell and imp producer of an aquatic form

Q:27 Define Omnivorous? (primary consumer)
Can eat everything: Includes pigs, foxes, coachroaches & Human. Dentition:- Their teeth are • ripping • chiseling • grinding

Q:28 Detritivous feeders?
That feed on wastes or dead bodies, known as detritivous feeders. e.g mites, earthworm
• Catfish • vultures

Q:29 Define Decomposers?
Decomposers break down dead organism & feces by digesting organic matter extracellularly & absorbing the product of digestion
They feed by secreting enzyme e.g bacteria & fungi

Q:30 Features of phytoplankton?
Microscopic organism that floats or drift in water. They a great source of food for aquatic creatures. Like: whales, shrimp, snails
They are 2 types: Diatom
Dinoflagellate

Q:31 Define Red tide?
The harmful & toxic bloom of dinoflagelle causes the colour of sea water to turn red e.g in River nile Egypt.

Q.32 Food web ?

In natural ecosystem most consumer feed on more than one type of organism, & most organism are eaten or decomposed by more than one type of consumer.

bcz of this, organism in most ecosystem form a complex network of interconnected food chain called food web, It shows how producer, consumer & decomposer are connected to one another as flow of energy through trophic level in ecosystem provide stability in ecosystem.

Q.33 Food chain ?

The sequence of organism through which energy moves in an ecosystem called food chain.

plant → insect → frog → hawk

decomposes use died hawk & return their nutrient to reuse of by produce.

Q.34 Food web ?

When an abiotic factor become out of an organism tolerance range then it is said to be limiting factor - e.g. Oxygen, water, food

Q.35 Taxis ?

Response of an organism towards any abiotic factor is called Taxis, an organism orient itself to response e.g response to light called photo taxis.

Q.36 Trophic level ?

Food web is complex, so its easy to group organisms according to the form of energy used, These group called ..... :- consumer; primary, secondary, tertiary

Q: 37 Producers :- photoautotroph :- chemoautotroph

producers are autotroph (self feeder) nutrient & energy obtained by environment

2 types :- photoautotroph & They captuse sunlight to produce energy rich carbohydrate (glucose) by photosynthesis

chemoautotroph

Bacteria convert inorganic from environment energy through complex nutrient compounds by using hydrogen sulphide instead of sunlight called chemosynthesis

Q: 38 Ecological Niche & role of bears in ecosystem?

Ecological niche is the role of animal in our ecosystem. It is defined as the role that specie plays in environment including behaviour & influence. Its a way of life in a comunity that includes everyting that affect its survival & reproduction -

- activity pattern: remain active
- actual place of living
- resources gain from surrounding environment. [role of bears in ecosystem]
- Lives in forest
- feeds on nuts, blood sucking insects & parasites.
- carry seed of berries in stomach for long distance & expel in new place to grow in new area
- hybernate in winter season.

Q: 39 How living system are sustained?

All living systems from cell to biosphere are dynamic system that are constantly changing in response to changing environmental conditions.

* continent movement
* climate changing
* Natural disturbance
* Succession

living system contain complex networks of positive & negative feedback loops that interact to provide some degree of stability, or sustainability, expected life span.

Q: 40 Factor of carrying capacity?

The maximum population of a living specie that a particular habitat can sustain without being degraded - The carrying capacity of an area or volume is not fixed.

availability
- water
- space
- Environment condition

Q.41 Define weathering ? process of decomposition & disintegration that causes the rocks to break down in natural way i- Chemical, Mechanical Biological weathering

Q.42 Earth life support system ?
Earth life support system consist of four main physical spherical system that interact with one another
- atmosphere (air)
- The hydrosphere (water)
- geosphere (rocks/soil)
- biosphere (living thing)

Q.43 Importance of coral reef ?
- corals are tiny living animals that belong to a group of organism called Cnidarians. They are two main types of coral reefs - • soft corals • hard corals.
- coral reef produced by large mass of calcium carbonate that secreted by coral in marine water.
- They an amazing centre of aquatic biodiversity.

Benefits:-
- provide natural habitat • Economic benefit
- food • Medicine • Recreation • breeding ground • Nursing ground.

Q.44 Diff b/w rocky & sandy shores ?
The area of shoreline b/w low & high tide is called intertidal zone.

* Rocky shores :- They face a huge impact of waves. so large no of specie occupy diff niche in intertidal zone in response to following changing environmental condition.
- Temp. • water flow • salinity

- Sandy shores :- Sandy beaches & their adjoing coastal wetlands are a home to a variety of shore birds that feeds in specialized niche on crustaceans, insect & other organism. Organism can survive by adopting following adaptations
  - burrowing in sands • Digging in sand
  - Tunnelling in sands.

Q: 45 Soil Importance:- • Nutrient • water • habitat
• Seed germination • Biological productivity

Q: 46 Soil Type :- can be determined by • Topography
• climate • Time length -
In general diff b/w Two type of soil

1 -> Pedalfer Soils-
• Acidic soil formed in wet temp.
• Extensive leaching activity
• Support growth of deciduous forest & tall grasses-

2- Pedocoil Soil :- (acid & semid)
• Soil that formed in dry climate & basic in nature
• less leaching in activity
• promotes growth of desert shrubs & small / medium grasses.
↓ Types
3- Laterite soils-
a special type of pedocal that is bright red & form in Tropical & Subtropical climat with heavy rain & high Temp-

4- Caliche- A tough cemented layer developed by dense mass of calcium carbonate-

Q: 47 Soil Horizon-
A vertical section of soil that tells about various layers of soil as zones are named as Soil horizon-
It has 4 zones: O zone:- upper most zone of soil where leaves & fallen & other plant or mites deposited-
-> A zone Zone of leaching they accumulate in B zone
-> B zone Zone of accumulation they accumulate in C zone-
-> C:- composed of broken bedrock material that merged with underlying solid & hard bedrock material.

Q:40 Temperature?

Temperature controls metabolic rate as well overall activity of an individual.

Human: 98.6 F

goat: 103° F bird: 105° F

Warm blooded animal e.g. these animal that maintain their internal body temp. constant. Bids / mamals / homeotherms

Cold blooded animal e.g. animal whose internal body temp. changes accordance to external env't - Temp. also called Poikilotherms. Amphibians / fish / reptiles

Q:49 Imp of Temperature?

- Metabolism
- Seed germination
- opening & closing flowers
- Seasonal fallig of leaves
- Growth & development
- Animal migration due to seasonal changes in temp.
- reproduction - Temp - (hot day & cold night) in desert

Q:50 Define Thermoregulation?

Ability of an organism to regulate its body temp.

- Metabolism
- Growth
- reproduction

Influenced by

- Availibilty of food
- water
- Shelter

-> Optimum temp:- Enzyme a biological catalyst needs a specific temperature.

Improper temp. makes the enzyme unable to perform their catalytic activity.

Q: 51 Write down the concept of a specie?

Ans A group of sexually reproducing organism that can mate & produce fertile offspring. This is biological concept of specie.

- Every organism is a member of a particular specie with specific trait.

→ Biological concept of specie :-
This defines specie on the base of their interbreeding potential not on similarities or appearance. e.g western meadowlark & eastern meadowlarks look similar but they do not interbreed with each other, so they r separate species

→ Ecological concept of specie :-
It defines specie in terms of its ecological niche.

→ Morphological concept of specie :-
Its define specie by its structural features.

→ phylogenetic specie concept :-
define specie as the smallest group of individual on a phylogenetic tree.

Q: 52 Population :- A group of same specie located at the same time is called population.

Q: 53 Population characteristics :-

population size	population disturbance
population growth	population density

Q: 54 population limits :- If environment is bad place to stay or live it must be to hot, too cold, or too dangerous.

Geographical limit :-
it also a main constraint that decrease the size of popu

Q: 55 Community :- is the group of population of diff Specie located in a particular area or place.
- Communities are not only random composition of species but they also hv a unique organization.

Q: 56 Community Assemblage :-
The community is an assemblage of species population that occurs together in space & time.

Q: 57 Community assembly :- The specie that assemble to make up a community are determined by.
- dispersal constraint
- Environmental constraint
- Internal dynamics

Q: 58 Biotic components :- producer = green plants
consumer = Animal
Decomposer = bacteria & fungi

Q: 59 Generalistic Specie :- the species which have broad range of niches are known as generalistic species.
- they can live in many diff places, eat variety.
- often tolerate a wide range of environmental condition.
- Eg :- cockroaches, mice, rats, racoon & human

Cockroaches present are 350 million years
- high reproduction rate
- eat variety of food
- can survive after 45 to 60 days freeze
- resistant to poison
- plays role in food web, tasty meal for birds & lizards.
- about 25 species, female carry 25 diff viruses & bacteria in their

Q: 60 Specialist Specie :-
Those specie that occupy narrow range of niche are known as specialist species

- They r more prone to extinction-
- They may be able to live only in one habitat
- use one or few type of food
- Tolerate a narrow range of climate & Env. changes

e.g tiger salamanders :- breeds only in fishless ponds when their larva will not be eaten.

- They show reduce competition.
- They allow sharing of limiting resources.

Q[61]: Is it better to be a generalist or specialist?
Specialist hv an advantage bcz they hv few competitors but under rapidly changing environmental condition, the generalist can adopt simply so, they can survive it depends on the present condition.

Q[61]: Native Species are those species that normally live & thrive in a particular ecosystem bcz of its heat, isolation & climatic conditions.

- They occurs naturally with in a region.
- They either survive in that region or arrive & become established in that region without human assistance.

Endemic Species: These species (e.g indus dolphin in pk only) r found with in specific part of world only.
Indigenous Species: They r found with the region & elsewhere in world also.
Snow leopard in india & russia, pk etc. :- Red kangroo only in australia only.

Q[62]: Non-native Species-
Species that are non-native to their env. & are deliberally or accidently introduced into an ecosystem & called non-native species.

- Also named as invasive . alien . exotic species

- Invasive species are those who harmful to humans or other communities

Q: 63 Impacts of Non-Native Specie?

- Non-native specie spread rapidly if find new more favourable niche
- In new niches these specie often do not face the predators & diseases they face.
- They may be able to out-compete some native specie in their new niche
- Causes habitat loss & ecosystem disturbance

Q: 64 Indicator Specie?

Specie that provides early warning of damage to a community or an ecosystem called Indicator

- e.g. The presence or absence of trout specie in water at Temp. with in their range of tolerance is an indicator of water quality, cz they need clean water with high level of oxygenized -

Q: 65 Keystone Specie?

Some specie plays a key & crucial role in maintaining the structure of a Biological community - known as keystone specie - They have large effect on types & abundance of other specie in an ecosystem

- e.g: Sharks also a keystone specie bcz they remove injured, sick animals from ocean, thus they play an imp - Eco. role.

Q: 66 Fourdation Specie? good e.g. of beaver made dams on rivers & other wetlands to make pathy for other spec.

plays a major role in shaping communities. They create & Enhances the habitat in ways that provides benefits to other specie. They facilitate the survival & reproduction of other species in com

- e.g:- Elephants plush over, beetles, upsoat trees, forest openings in grasslands & wetlands of Africa which helps others.

Q: 67 Ecological pyramids ?

An ecological pyramid is graphical relationship among diff trophic structures or level in an ecosystem - • Trophic means diff levels of organism- 3 main type:

→ Pyramid nos - represents the no of individual at each trophic level of an ecosystem-

→ Pyramid biomass- Indicates mass of living matter at each trophic level of food chain

→ pyramid of energy - determines the flow of energy through each trophic level in food chain

Q: 68 Food Pyramid ?

It is a graphical representation of flow of energy through diff trophic level of eco.

(Population (v.imp))

Q: 69 Population Dynamics ?

It is a study of how these characteristics of population change in response to changes in environmental condition : Environmental conditions ar

→ Temperature → harmful chemical
→ resources availability → driving or disapering of competing spe

Q: 70 Population dispersion (distribution): 1. clumping
2. uniform clumping 3. random clumping

Q.71 Population factors:

- Distribution
- Numbers

Population differ in factor such as
- age structures
- Density

Q.72 clumping:-

Individual s found in form of groups or clusters due to uneven distribution of resources.

Q.73 Population / uniform dispersion:-

the tendency for population to be found evenly distributed about their habitat

- This is caused by a specie ability to survive anywhere in their habitat

Q.74 Random / Population dispersion:-

distribution of population in random fashion through out their habitat generally, this type is rare while clumping distribution is common.

Q.75 Population growth:

Animal population change over time as result of

- Birth
- Death
- Dispersal

→ Biotic potential: Ability of population to grow under ideal condition

→ low biotic potential Large individual, such as elephant & blue whale

→ High biotic potential Small individual, such as bacteria & insects.

Q: 76 Intrinsic rate of increase :- the rate at which population of specie would grow if it had limited resources.

- response early life
- short generation time
- reproduce many times
- how many offspring may live

Q: 77 Environmental resistance :-

the constraints that climate, food, space, & other environmental factors place on a population called Env. res. -

Q: 78 Carrying capacity K ?

It is combination of all factors that act to limit the growth of a population - Together, biotic potential & environmental resistence determines the carrying capacity K-

Q: 79 Survivorship curves : plots of No. of survivors -

Type 1 curve : live out their potential life span

" 2 " : mortality rates are constant throughout age classes

" 3 " : mortality rate is higher for youngest cohorts.

Q: 80 Define Exponential growth ?

A rapid increase in no. of indiv of a population at a constant rate of growth

→ Exponential or geometric growth starts slowly but then accelerates as the population increases, bcz the base of size is increases.

Q.81 Population Age Structure :- has a strong effect on population growth. Populs usually discussed with:-

→ Pre-reproductive age :- Not mature enough to reproduce
→ reproductive age :- Those capable to reproduce
→ post reproductive age :- Those too old to reproduce

Q.82 Reproductive potential :-
Ability of individual to reproduce under ideal or optimum condition. Exponential growth cannot occur indefinitely.

Q.83 Factors of reproductive Influence.
→ No. of offspring produced
→ The likelihood of survival to reproduce age
→ Duration of reproductive period.
→ Length of time it takes to reach maturity

Q.84 Logistic population growth.?
Logistic growth involves rapid exponential potential growth followed by a steady decrease in population growth until population size is level off.

Q.85 Population regulation :-
Demography :- Study of human population & Factors that regulates the growth of population

Q.86 → Factors regulating Population :-
- Birth rate
- Death rate
- Desease
- Food
- water
- climate
- Disease
- competition
- Emigration

Q.87 Metapopulation: A group of population of same specie which is seperated by space - Individual can move from one population to othe - Extinction & colonization are the main proces that regulate metapopulation.

Q.88 Population Density:
↓ is the no of ind- in population found in particular area

Q.89 Higher population density:-
- May help sexually reproducing individual find mate but it can also lead increase compt. for mates, food, living spaces, water, sunlight
- Help to sheild some member from predators
- close contact among individual in dense population can increase the transmission of parasites & infectious disease
- Smaller organism have higher popul. dens than larger organ.

Q.90 Low population density:-
- less comp. for food, living space, water, sunlight other resour.
- they find diff. to find mates
- less adaptability to env. changes.

Q.91 Density Independant factors:-
Factors that influence the no of animals in population & their effect is not dependant on the density of the popula. So, D.I exerts a strong effect on popula. size.

Q: 92 Population dispersion: or distribution patterns show the spatial relationship b/w members of population within habitat.

Q: 93 density Independent factors:

- Weather
- Floods
- Fire
- climate changes
- other factor of climate
- late Spring
- Earthquake
- Hurricane
- Pollution
- Habitat destruction
- Severe freeze

Q: 94 Factors that influence population density

- Comp.
- Disease
- Predation
- parasitism
- food
- migration

Q: 95 Population Integrity?

it refers to the independence or discreteness of subpopulations. Therefore, population complexity is the result of trade-off b/w connectivity & integrity.

Q: 96 Impacts of human population?

H.P increases day by day about 74 billion
Increases in population causes in increase of demand/use of Nat. resources
It affected the Env. by putting pressure on natural resources

- Habitat destruction
- Habitat fragmentation
- Pollution
- Soil erosion
- Invasive Species
- Oil Spills
- Coral reef distrctn
- Ozon layer dest.

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- Today my zoo507 paper.
- Mcqs:
- 1. Living and non. Living things includes ---- (invironment).
- 2. Non. Renewable sources such as aluminun and _____.
- 3.comunity of different ___ interact each other.
- 4.organisms are divided into ____ types.
- 5.viola triclor live in ___ environment.
- 6.ecological niche is important for the study of___.
- 7.blind indus dolphen belong to ____ specie.
- 8._____process start ecological food web.
- 9.% age of nitrogen in atmosphere?
- 10. _____Factor helpful to regulate the population?
- 11.%age of water that cover the earth surface.
- 12.____ zone contain mountain head water streams.
- 13. Ectotherm organisms regulate their body temperature through____.
- 14._____ wave length of electromagnetic radiations detect human eye.
- 15. Tropical rain forest contain ___ of total world life.
- 16.competition among same spacies are called_____.
- 17.commensalism.
- 18.foliose is growth form of _____.
- 19.camouflage.
- 20.grass land biomes is example of _____.
- 21.E1 Nino effect occurs in _____ years.
- 22.CFCs stand for.
- 23.the %age of plants and animals spacies on world.
- 24._____ is basic unit of life and responsiple for the trait inherited.
- 25.pollutants.
- 26. Competition within a specie member.
- 27.PAR stand for_____.
- 28.physical location of gene are _____.
- 29.mRNA is furthur translated into _____.
- 30. When extra neucleotide put into DNA is called_____.
- 31.human affected environment _____ %.
- 32.invasive specie are called _____.
- 33.hibernation in large animals known _____.
- 34.aposematic coloration.
- 35.sulphur circulate through _____ cycle.
- Subjective.
- 1.aestivation?2 marks
- 2.solid waste ? 2marks
- 3.exploitation? 2 marks

- 4.growth forms of lichens? 3 marks
- 5.acid rain? 3 marks.
- 6.three sources of energy?3 marks
- 7. Three process involve in water cycle? 5 marks
- 8.compitition and difference between interspecific and itraspecific .? 5 marks
- 9. Primary succession with examples? 5 marks.
- 10.factors responsible for genetic diversity of a small popultion? 10 marks

Zoo 507

→ How plants converted light energy into chemical energy?

(Specie)

→ Foundation specie e.g
→ concept of specie e.g
diff → b/w Indicator specie e.g → foundation specie
→ keystone specie e.g
→ Non-active specie e.g
→ How specie diversity affect ecosystem e.g
→ How non-native specie affect on environment e.g
→ Diff b/w generalist & specialist specie e.g
specie diversity ? → Indicator specie ?

(population)

→ reproduction potential
→ population dynamics → factors of population dynamics?
→ population integrity
→ population dispension
→ charcter of population
→ population depression

(Environment)

→ effects of human on environment ? → factors are affected by community
→ ecological footprints are growing
→ Inlist composition of atmosphere in ecosystem
→ Eutropication
→ Imp of water H_2O ?
→ composition of geosphere :- earth science
→ Major factors that support life on earth